Artificial Intelligence in Healthcare

Project title: AI-Powered Clinical Decision Support System Using Reinforcement Learning

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Background

- In real life, doctors use experience and clinical knowledge to make these decisions, but building an AI system that can assist with this process can improve efficiency and support decision-making.
- Since real medical datasets are difficult to access due to privacy and ethical concerns, I used synthetic patient data to simulate real-world cases. This project focuses on using Reinforcement Learning (Q-learning) to recommend treatments, suggest doctor types, and generate explanations with audio support for better user interaction.

Goal

- Build an AI system that:
- generates patient cases
- predicts disease from symptoms
- selects the best treatment
- suggests the correct doctor
- explains the decision using LLM
- provides audio explanation

Project Objective

Use Reinforcement Learning (Q-learning) to help simulate clinical decision-making.

Compare with:

- Random policy
- Rule-based policy
- RL policy

Final Goal:

Check whether RL can learn treatment decisions close to expert rule-based logic.

Synthetic Patient Generation and RL Environment Design

Why Synthetic Data?

- Real medical datasets are difficult to access.

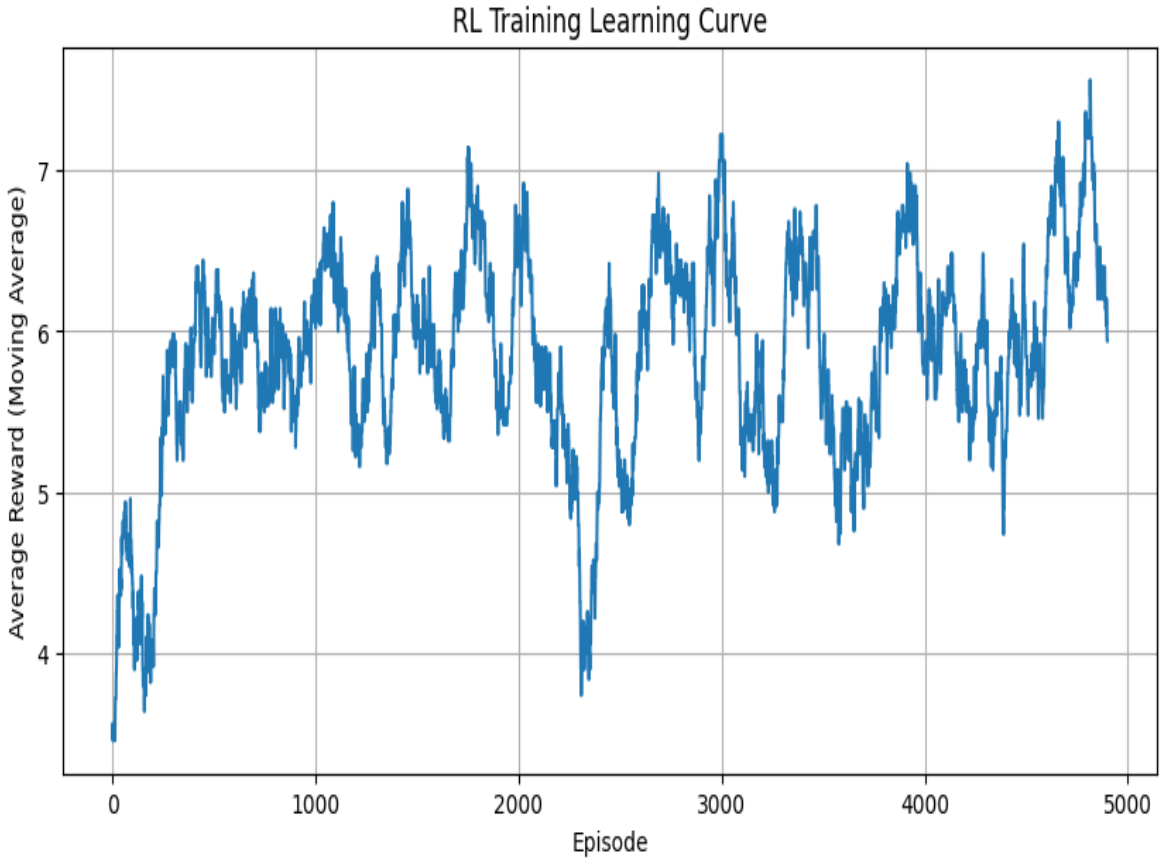
What I generated

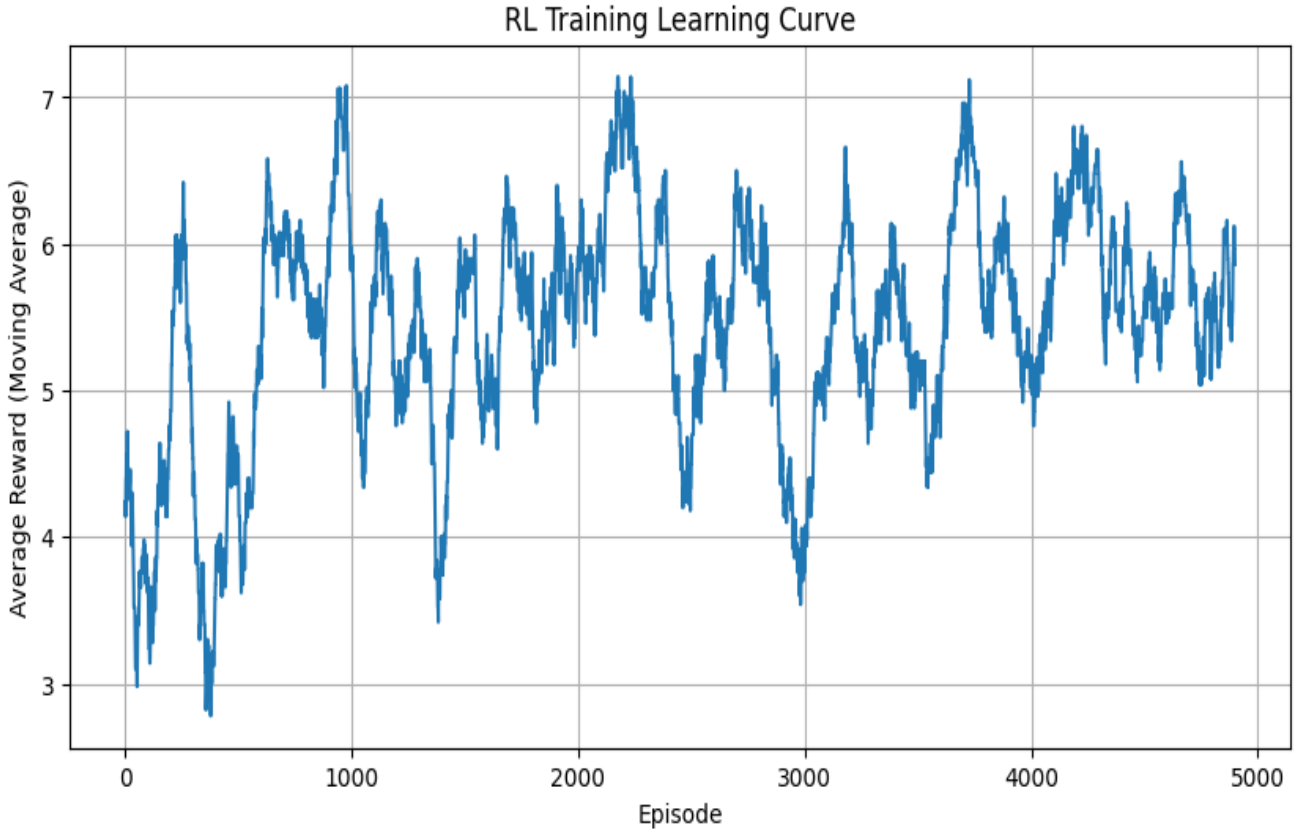
- Age
- Symptoms
- Severity
- Disease type

RL Environment Design

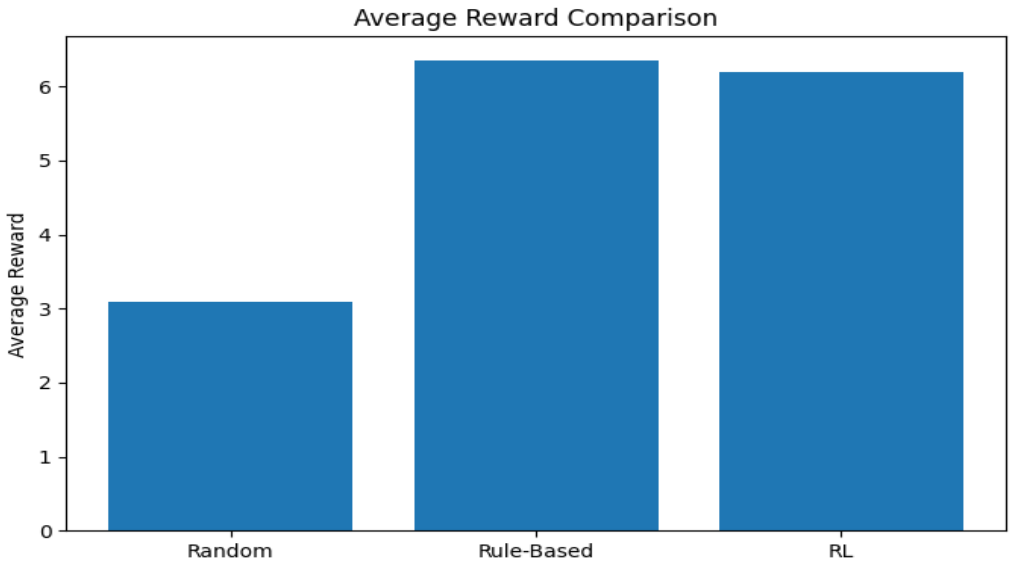
Environment = Clinical TrialEnv
State
(Disease, Severity Level)
Example:
(Covid, High)
Actions
Treatment A / B / C
Reward
Improved = +10
No Change = 0
Side Effect = -6

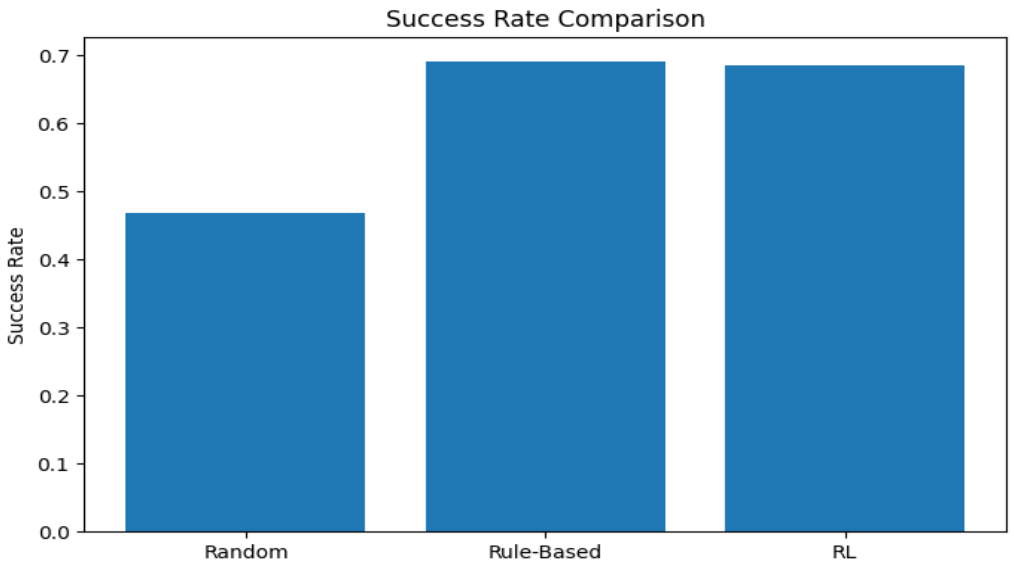
Q-Learning Curve



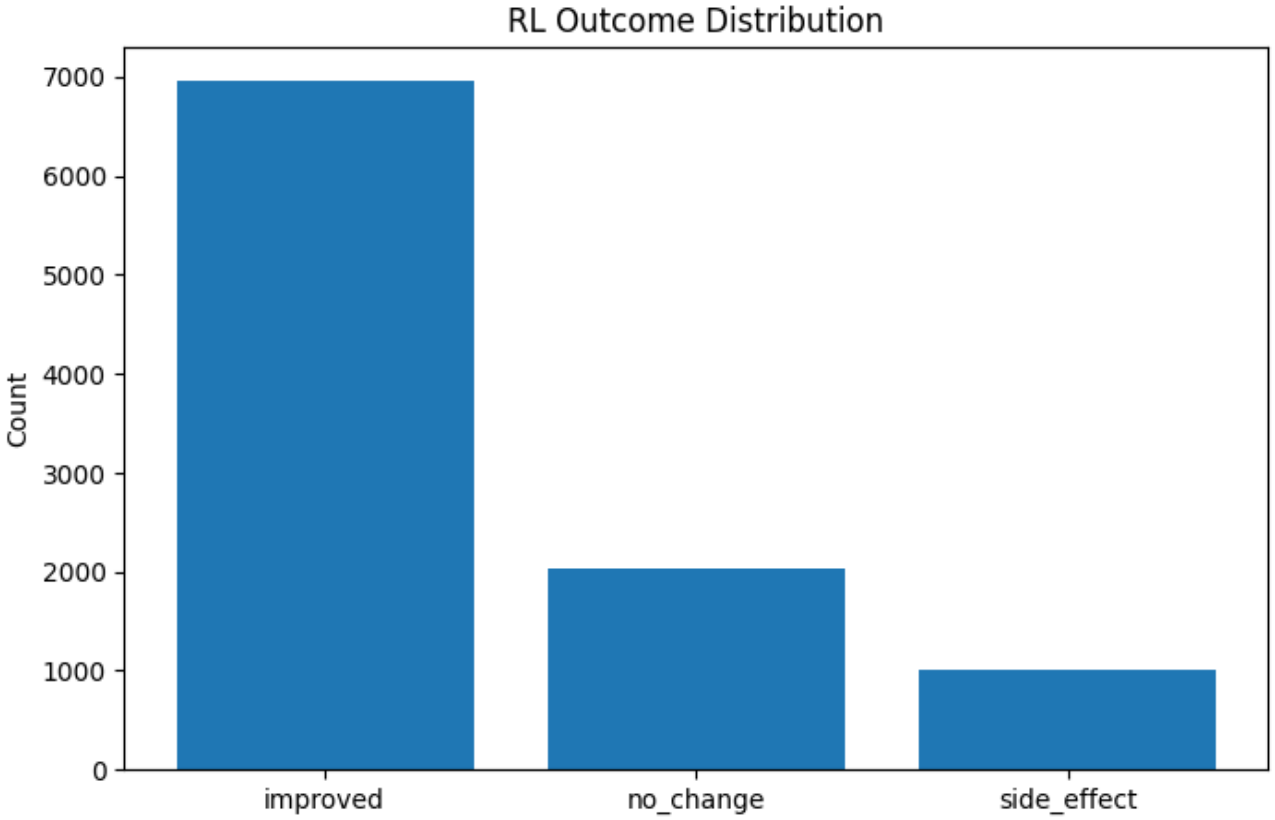


Policy Comparison Results





Results



LLM Explanation + Audio Feature on Result

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Case 1
----------------------------------------
Age: 69
Disease: Hypertension
Severity: 9 (high)
Symptoms: {'fever': 1, 'fatigue': 1, 'cough': 0, 'chest_pain': 1, 'high_bp': 1, 'shortness_of_breath': 0, 'loss_of_taste': 1, 'high_sugar': 0, 'frequent_urination': 1}
Chosen Treatment: Treatment B
Suggested Doctor Type: Cardiologist
Outcome: improved
Reward: 10

Explanation:
Please remember, this is a simulated educational system and not real medical advice.

Given the patient's age (69), severe hypertension (severity 9), and critical symptom of chest pain alongside high blood pressure, a Cardiologist is the most appropriate specialist.
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more options

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Case 2
----------------------------------------
Age: 18
Disease: Flu
Severity: 10 (high)
Symptoms: {'fever': 1, 'fatigue': 1, 'cough': 1, 'chest_pain': 1, 'high_bp': 1, 'shortness_of_breath': 0, 'loss_of_taste': 0, 'high_sugar': 0, 'frequent_urination': 0}
Chosen Treatment: Treatment A
Suggested Doctor Type: General Physician
Outcome: improved
Reward: 10

Explanation:
Please remember, this is a simulated educational system and not real medical advice.

In this simulated case, a General Physician is suggested because the flu, even with high severity and symptoms like chest pain and high blood pressure, is a common illness that GPs
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Difficulties Faced and Future Plan

Difficulties

- Real medical data was difficult to access
- Creating realistic synthetic patients was challenging
- RL training had fluctuations and debugging issues
- Gemini API had quota and model errors
- Audio integration required handling empty explanations
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Future plan

- Use real hospital datasets
- Add multi-step treatment decisions
- Improve prediction with more ML models
- Make the app more interactive

Conclusion

This project successfully built an AI-based clinical decision support system using Reinforcement Learning. The system generates synthetic patient cases, predicts disease, recommends treatment, and suggests the appropriate doctor. The RL model performed much better than random policy and close to the rule-based approach. Adding LLM explanations and audio support made the system more interactive and easier to understand. This project shows how AI can support smarter healthcare decision-making in a simulated environment.

Thank You 😇

“Stay safe, stay healthy, and let AI help build a better healthcare future.”